

SHIVAM TAMBOLI

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TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript
- **Backend Frameworks:** FastAPI, Spring Boot, Spring AI, Spring Data JPA, Hibernate, Express.js, Node.js
- **AI / Backend Systems:** Retrieval-Augmented Generation (RAG), Vector Search (HNSW), LLM Integration, Hybrid Search (BM25 + Semantic), Reciprocal Rank Fusion, AST Parsing, Async Processing, Background Task Processing, Server-Sent Events (SSE), Cache-Aside, TTL Caching, Rate Limiting, JWT Authentication, Role-Based Access Control (RBAC)
- **Databases & Infrastructure:** PostgreSQL, MongoDB Atlas, Redis, MySQL, Docker, Alembic, Maven, Git
- **Frontend :** React, Next.js, Tailwind CSS, HTML, CSS

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Siddhesh Private Limited Technologies

Feb 2026 – June 2026

Learning Management System

Next.js, Node.js, TypeScript, PostgreSQL, Express.js, JWT

- Built a RESTful backend with Express.js and PostgreSQL implementing JWT authentication, role-based access control (Admin/Student), and modular APIs for authentication, course management, and student workflows.
- Engineered normalized relational schemas with typed, ESLint-enforced TypeScript across the server layer; delivered a supporting Next.js + Tailwind CSS frontend with role-based routing, while primary focus remained on backend API design and data modeling.

PROJECTS

AI Codebase Assistant

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Backend: FastAPI, Python, MongoDB Atlas, OpenAI API, Cohere, Docker, JWT, pytest | Frontend: React

- Designed and built the complete backend: RESTful API (FastAPI), async ingestion supporting ZIP upload and GitHub import (public + private), hybrid search engine, SSE-streamed LLM answers, and user-scoped JWT authentication with per-IP rate limiting — structured as independent, testable service modules.
- Built a multi-language AST parsing pipeline (Python `ast`, tree-sitter for JS/TS/Go/Java/Rust/Ruby) with symbol-aware chunking at function and class boundaries; implemented MD5 hash diffing for incremental re-indexing to eliminate redundant embedding API calls on unchanged files.
- Architected a hybrid retrieval system fusing OpenAI semantic embeddings (`$vectorSearch` HNSW on MongoDB Atlas) with MongoDB `$text` keyword search via Reciprocal Rank Fusion ($k=60$), re-ranked by Cohere cross-encoder with BM25Okapi fallback.
- Designed a provider-agnostic LLM layer (Factory + Strategy) enabling OpenAI/Anthropic model switching via a single env variable; enforced citation validation by regex-matching cited file paths against retrieved chunk metadata; built a 41-test pytest suite with full mock isolation.

URL Shortener

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Backend: FastAPI, Python, PostgreSQL, Redis, SQLAlchemy (async), Alembic, Docker | Frontend: React, Vite

- Designed and deployed a full-stack URL shortener handling URL creation, redirection, and click analytics — backend on Render (FastAPI + PostgreSQL + Redis), frontend on Vercel; built on fully async SQLAlchemy with Alembic migrations configured for async engines.
- Implemented a cache-aside pattern with Redis to serve repeat lookups in sub-millisecond latency, falling back to PostgreSQL on cache miss and warming the cache on write so even the first click avoids a database read.
- Used FastAPI `BackgroundTasks` to increment click counters after the redirect response is sent — keeping redirect latency unaffected by write operations; debugged async session lifetime by giving background tasks their own independent database sessions.

LyricMind — AI Song Recommender

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Backend: Spring Boot, Java, Spring AI, MongoDB Atlas Vector Search, OpenAI API, Caffeine Cache, Docker | Frontend: React

- Engineered the full backend pipeline in Spring Boot with Spring AI: CSV ingestion parses song metadata, generates OpenAI embeddings (`text-embedding-3-large`, 3072 dims), and bulk-indexes vectors into MongoDB Atlas's HNSW vector index.
- Designed the retrieval and reranking layer to issue semantic `$vectorSearch` queries (cosine similarity threshold 0.6), then rerank candidates with GPT-4o-mini via a “select top N” prompt — keeping LLM output $O(\text{limit})$ rather than $O(\text{topK})$ for predictable latency.
- Added a Caffeine in-memory cache keyed on (mood, limit) with 15-minute TTL to eliminate redundant embedding and LLM calls on repeat queries (sub-5ms cache hits); implemented graceful fallback to raw vector-search order on rerank failure.

EDUCATION

MIT World Peace University — Master of Computer Applications (MCA)

2024 – 2026

Courses: Advanced Java, Web Technologies, Database Systems

Manipal University Jaipur — Bachelor of Computer Applications (BCA)

2021 – 2024

Courses: Data Structures & Algorithms, Object-Oriented Programming, Java

CERTIFICATIONS

Java Development – Coursera

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Generative AI Fundamentals – Coursera

[View](#)